

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE
OR BRAND NAME

PICK ONE

Sole Source:

☒ I recommend that NASA, NASA Shared Services Center negotiate with DS Government Solutions Corp only for PowerFLOW License Renewal. The total estimated cost of this effort is [REDACTED] and the estimated period of performance or lead-time for delivery is 4/6/2026 - 4/5/2027.

OR

Brand Name Justification:

☐ I recommend that NASA, NASA Shared Services Center negotiate only with vendors who can provide the subject band name from Manufacture Name for Enter Supply or Service to be procured. The total estimated cost of this effort is Enter Dollar amount and the estimated period of performance or lead-time for delivery is Date.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacturer. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

This procurement is to renew licenses that are being used currently by the Advanced Measurements and Flow Physics Branch to perform research in support of numerous NASA missions. PowerFLOW software is used to simulate unsteady flows over complicated geometries without the requirement for extensive and costly grid generation. This software suite includes the capability to rapidly generate embedded Cartesian grids with varying resolution. Proprietary technology for near-wall treatment allows the use of wall functions instead of resolving the flow all the way to the wall, reducing the grid requirement. Furthermore, the unique algorithm is very efficient on modern CPUs and parallelizes efficiently to many cores. The PowerFLOW suite also includes complete flow analysis and noise prediction codes that are required to fully analyze a configuration. Researchers in the Research Directorate (RD) have been running PowerFLOW simulation studies to investigate the aeroacoustic performance of the high-lift configuration of the Common Research Model (CRM). PowerFLOW has algorithmic advantages that allow meshes to be quickly generated around complicated geometries and can perform unsteady simulations with acoustic predictions in a relatively efficient manner. These advantages result in a better turnaround of simulations than our other tools, which will allow more configurations to be evaluated before the wind tunnel test.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

Our branch members have developed a level of proficiency using the PowerFLOW software suite which results in an efficient workflow. Switching to another brand would result in delays and inefficiencies in production. No other product is compatible with PowerFLOW datasets, and Dassault Systems is the only entity authorized to sell the software in the United States. The introduction of a new brand of software to meet this functional requirement would have repercussions on project timelines, budgets, and staff.

ADDITIONAL COMMENTS: